

Y. BENJAMIN PEREZ C.

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PROFESSIONAL SUMMARY

Postgraduate Student in MSc. **Artificial Intelligence**, with BEng. in Computer Systems Engineering, with over 4 years of hands-on programming experience through academic projects, internships, and self-led development, with programming knowledge of **Python, C++, and Java**, from different academic projects, internships, and freelance development. s.

SKILLS

Programming Languages: Python, Java, C, C++, MATLAB

Frameworks & Libraries: TensorFlow, Pytorch, OpenCV, Numpy, Pandas

Data & AI: Convolutional Neural Networks (CNNs), ARIMA, MySQL

Tools & Platforms: Git/GitHub, Google Cloud Platform (GCP), Amazon Web Services (AWS)

Languages: English (fluent), Spanish (fluent), Mandarin (beginner)

EDUCATION

BEng. (Hons) Computer Systems Engineering Lv. 6

Brunel University of London | Sep 2022 – Jun 2025 | Grade: 2:2 (Lower Second Class)

MSc. Artificial Intelligence Lv. 7

City St. Georges' University of London | Sep 2025 – Jun 2026 / 27 (with Placement) |

- Principles of Artificial Intelligence
- Programming and Mathematics for Artificial Intelligence
- Computational Cognitive Systems
- Agents and Multi-Agent Systems
- Deep Learning for Image Analysis
- Deep Learning for Sequence Analysis
- Explainable Artificial Intelligence: Ethical and Legal Challenges
- Deep Reinforcement Learning
- Individual Project

CERTIFICATIONS

- Finance Fundamentals with CFI - **DataCamp** (in Progress) | 2025
- AI Fundamentals – **DataCamp** | View Certificate | 2025
- Fundamentals of Process Mining – **Celonis Digital Academy** | View Certificate | 2025
- Introduction to Quantum Computing – **UCL** (Future Learn) | View Certificate | 2024/25

PROJECTS

ECG Signal Data Pre-processing | Source Code | **Python, Keras, CNN, OpenCV, Numpy, Pytorch**

- Implemented a pipeline to convert scanned ECG readings into structured time-series data using preprocessing (background removal, thresholding, Gaussian blur) and edge detection (Canny + contour extraction). Extracted (x,y) pixel coordinates per column, ordered them into time-series traces, and exported to CSV for downstream analysis.
- Applied Pandas/NumPy to clean and reshape large pixel matrices and used Matplotlib to validate continuity and visualize signal morphology.

Machine Learning based Occupancy Monitoring System | Source Code | **Python, TensorFlow, CNN (Neural Networks)**

- This project developed a contextual AI system using Python and TensorFlow, implementing lightweight Tiny-ML models on Edge Computing with a minimal dataset strategy for ML, to optimize office environmental control.
- The approach leveraged TensorFlow-based Tiny-ML for real-time, localised data processing, enhancing energy management and operational performance.

Dot-Tracking (Raspberry Pi) Camera Project | Source Code | **Python, OpenCV, Wireshark Client**

- Using Raspberry Pi(s), (specifically Model 4B) to detect red-dot objects, and measure distance travelled (i.e. motion in x and y coordinates) through streaming video footage from a Remote RPi and capture it from a Local Device sharing the same Wi-Fi network.
- Connected to a Multi- access Edge Cloud Linux Instance, i.e. VM on Google Cloud Platform, and x2go Client for GUI interfacing.

Banking / Supermarket Management System | Source Code | Java, SWING Framework, MySQL

- Project using different classes and methods to create a Banking Java Program implemented with a Supermarket Software System to store the personal details of the cardholders and serving customers using a digital ecommerce “online shopping system”.
- Created UML relationships between classes and objects, e.g. Bank Manager access, and other UML diagrams such as Sequence, use case diagrams etc. Used Object Oriented Programming principles, e.g. Polymorphism, Encapsulation, Inheritance, Instantiation, Abstraction.

EXPERIENCE

04/2024 – 07/2024 | **Systems Design Engineer Intern** | HTPM Research Project, Brunel University of London | *Relevant*

- Engineered control panels for heat pumps and energy systems, focusing on secure and efficient system design, including the preparation of Process and Instrumentation Diagrams (P&ID).
- Managed inventory of tools and equipment, ensuring all assets were accounted for and securely maintained, by utilizing Python-based workflow for inventory tracking, integrating image upload/resizing (OpenCV, PIL) with SQLite database storage, alongside Excel for reporting and asset verification.

06/2021 - 08/2021 | **Junior WordPress Developer Intern** | Complex Creative (Digital Marketing Agency) | *Relevant*

- Mentored and trained to implement WordPress websites from design concepts given by the client/s specifications, and collaborated with Digital Marketing and Graphic Designers, to help create a website using WP-Bakery plugin.
- Stayed up to date with new web technologies and emerging industry trends, by enrolling in online certification courses, in Security and eCommerce for WordPress plugins/websites.

05/2023 - 07/2023 | **Brunel Career Ambassador** | Professional Development Centre (Brunel University of London)

- Worked with the Professional Career Development (PDC) Team, by assisting public events through giving out flyers/help setup banners.

01/2022 - 08/2022 | **Sales Team Member** | Clarks | (Oxford Circus, Greater London)

- Helped assist customers and worked as a team to solve problems in the stockroom, and upfront instore. Totalled bills using math skills and maintained accurate till count.

ACHIEVEMENTS & EXTRACURRICULARS

Acted as **Course/Program Representative** for *Computer Systems Engineering* | 2024/25, and *Artificial Intelligence* (Current)

Recipient of 1st Prize Award for **Endless Blockchain Track**, Brunel University, Blockchain Society Hackathon | 2024

Part of the winning team selected by Endless and Luffa during Brunel Blockchain Society's hackathon.

Graduate of the **READY Programme** (Brunel University of London) | 2023/24

Graduate of the **Professional Mentoring Scheme** (Brunel University of London) - invited to Goldman Sachs's office | 2023/2025

Graduate of the **Graduate Aspiration Programme** (Civil Service) | 2024/25

Recipient of the Brunel+ **Bronze Award** (Extracurricular Events) | 2023/25